DRM Group Project

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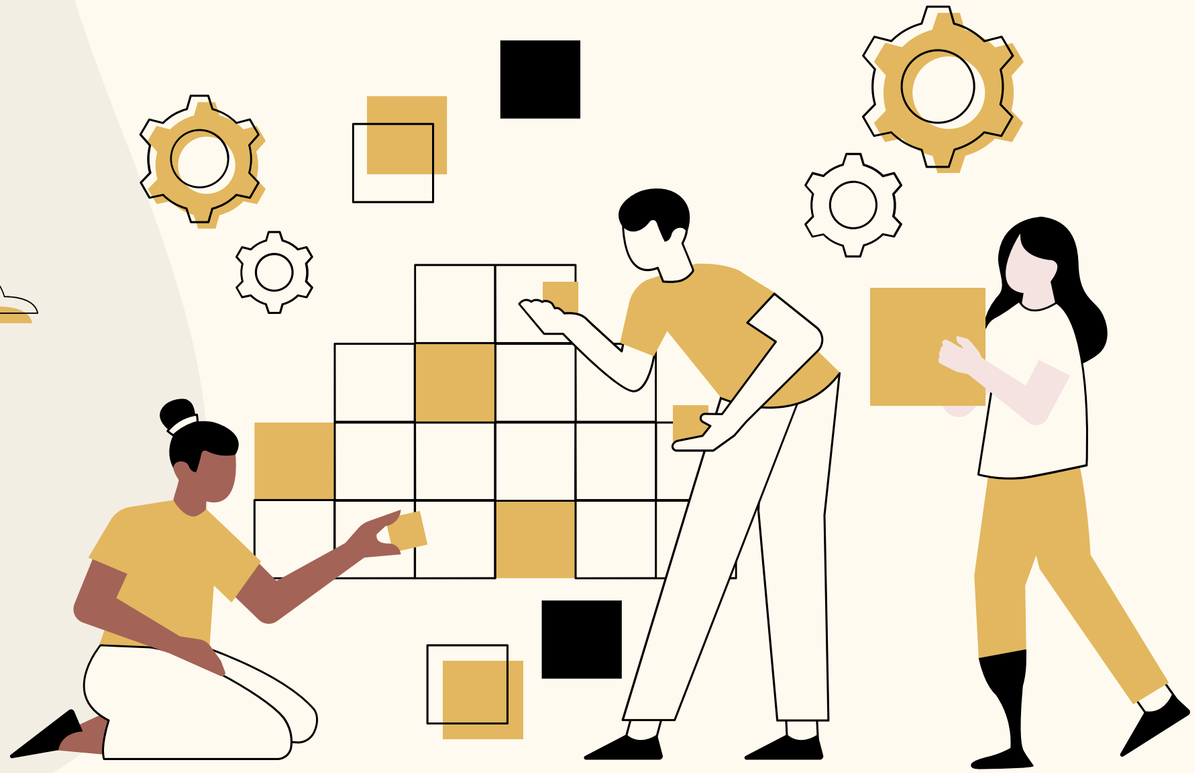


Task A - Synthetic Option Portfolio (Original & Alternate Versions)

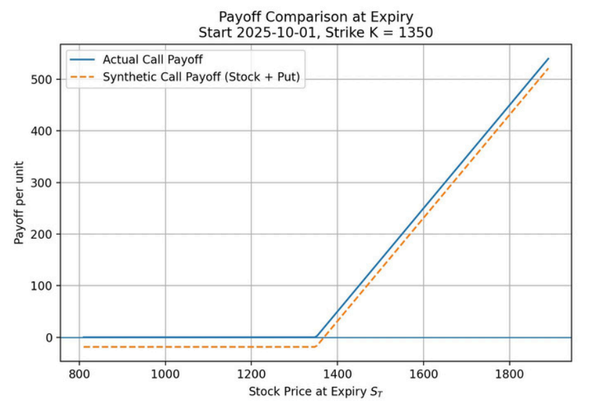
Actual vs Synthetic Call + 2-Year Strategy Backtest

A. Mandatory– Synthetic Long Call (Put–Call Parity)

KEY NUMBERS

Strike K = 1350

- Actual Call Premium: ₹46.23
- Put Premium: ₹17.93
- Synthetic Cost: ₹1,386.63
- Actual Call Cost: ₹46.23



Analysis Report

Constructed a synthetic Long Call = Long Stock + Long Put (ATM, same expiry) and compared it to the actual ATM call.

Key Insight

While put-call parity guarantees the same payoff at expiry, the two positions behave very differently in real markets because of practical frictions.

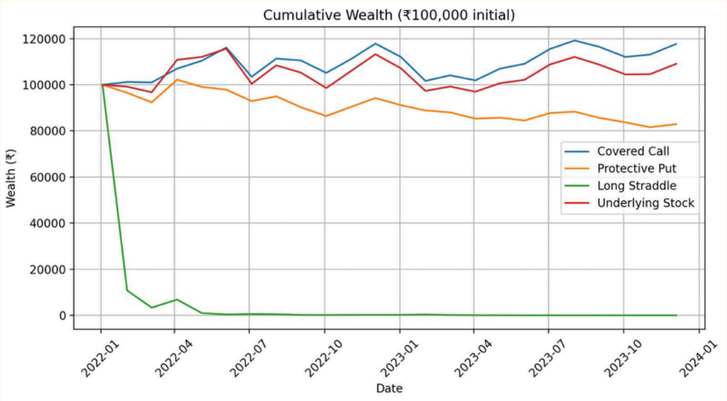
Reasons for Discrepancies in Synthetic and Actual Call

- **Much higher capital required:** Buying the full stock position makes the synthetic far more expensive and creates a big opportunity cost.
- **Different reaction to volatility:** The actual call benefits more from volatility (vega and gamma effects), whereas the synthetic moves more like the underlying stock.
- **Volatility skew/mispricing:** Calls and puts often trade at slightly different implied volatilities, causing small deviations from theoretical parity.
- **Liquidity + spreads:** Options, especially, puts tend to have wider bid-ask spreads, so executing the synthetic isn't frictionless.
- **Financing and carry costs:** Holding the stock ties up capital and carries financing costs that the actual call does not.

Conclusion

Although payoff matches at expiry, the actual call is more capital-efficient, responsive, and practical, while the synthetic underperforms due to real-world frictions.

A. Alternate – Options Strategy Backtest (2 Years)



Basic Idea

We backtested four option strategies Covered Call, Protective Put, Long Straddle, and Underlying Stock, over 24 months with ₹100,000 capital to understand how each behaves across different market conditions.

Key Insight

Each strategy reacts differently depending on whether the market is rising, falling, or moving sideways. The cumulative wealth results clearly highlight these regime dependent outcomes.

Market Regime Analysis (Bull, Bear, Sideways)

- **Bull Phases:** Covered Call and Underlying stock deliver the strongest performance as prices trend upward. The Straddle underperforms because volatility remains too low to offset time decay.
- **Bear Phases:**The Protective Put effectively cushions losses. Covered Call and Underlying decline similarly, while the Straddle reacts slightly better during sharp drops but stays deeply negative overall.
- **Sideways / Low-Volatility Phases:** Covered Call stands out, steady premium income drives returns even when prices don't move. Straddle collapses due to time decay, and Protective Put gradually loses value from repeated insurance costs.

Conclusion

Covered Calls excel in stable or range-bound markets, hedging strategies protect value in downturns, and long-volatility strategies like the Straddle only succeed when volatility is both high and persistent.

Task C - Binomial Option Pricing Model

Objective

Price European Call and Put options using a multi-step Binomial Tree and assess convergence with the Black–Scholes Model (BSM).

Key Inputs (User Provided)

- Spot Price (S_0): 1486.40
- Strike Price (K): 1500
- Time to Maturity (T): 2 years
- Risk-Free Rate (r): 6%
- Volatility (σ): 0.2125
- Steps in Binomial Tree: 400

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Model Output

- European Call Premium: ₹250.33
- European Put Premium: ₹104.68

Convergence Analysis (Binomial vs BSM)

- **Same foundation:** Both Binomial and BSM assume constant volatility and risk-neutral pricing; they differ only in how time is modeled.
- **Discrete vs continuous:** Binomial builds option prices step-by-step, while BSM works in continuous time.
- **More steps → better accuracy:** As the number of steps increases, the binomial price stabilizes showing the model has converged.
- **Converged prices match BSM:** At high step counts (N = 400), Binomial call and put values become very close to their Black-Scholes estimates.
- **Intuition:** With enough steps, the binomial tree essentially behaves like the Black-Scholes model.

Conclusion

The binomial model produced stable, theoretically consistent option prices, confirming that with sufficient steps, Binomial and BSM converge to nearly the same values for European options.

Thank You